

1. Why Probabilistic Reasoning?

- Many real-world domains are uncertain, noisy, and partially observable.
- Deterministic (true/false) reasoning is often too brittle.
- Probabilistic reasoning allows agents to represent and reason with uncertainty.
- Combines prior knowledge with new evidence to make rational decisions.
- Widely used in perception, robotics, language, medical diagnosis, etc.

Key Idea: Probability represents degrees of belief in uncertain worlds.

Example:

- A medical test is noisy (not 100% accurate).
- We still want to assess the probability that a patient has a disease given a positive test result.

4. Joint, Marginal, and Conditional Probability

- Joint probability: $P(X, Y) = P(X \wedge Y)$ probability that both events occur.
- Marginal probability: $P(X) = \sum_y P(X, y)$ (sum over all values of Y).
- Conditional probability: $P(X|Y) = P(X, Y) / P(Y)$ (if $P(Y) > 0$).

Example: Joint and marginal probabilities

Let E = Exercise (T/F) and R = Rain (T/F).

$P(E, R)$	$R = T$	$R = F$	Total
$E = T$	0.1	0.2	0.3
$E = F$	0.4	0.3	0.7
Total	0.5	0.5	1.0

- $P(E = T) = 0.3$ (marginal)
- $P(R = T) = 0.5$ (marginal)
- $P(E = T | R = T) = 0.1/0.5 = 0.2$ (conditional)

2. From Uncertainty to Probability

- Uncertainty arises from:
 - incomplete information
 - noisy observations
 - inherently random processes
- Probability provides a mathematical framework for quantifying uncertainty.
- A probability model specifies a set of possible worlds and a degree of belief for each world.
- Probabilities obey the axioms:

- $0 \leq P(A) \leq 1$ (for any event A)
- $P(\text{True}) = 1$ (probability of a certain event)
- $P(\text{False}) = 0$ (probability of an impossible event)
- If $A_1, \dots, A_n$ are mutually exclusive, $P(A_1 \vee \dots \vee A_n) = \sum_i P(A_i)$

5. Bayes' Rule

- Bayes' rule relates posterior and prior probabilities using evidence.
- Often used to update beliefs after observing new information.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- $P(A)$: prior probability (before seeing B)
- $P(A|B)$: posterior probability (after seeing B)
- $P(B|A)$: likelihood (how likely evidence B is if A is true)
- $P(B)$: evidence probability (normalizing term)

Example: Medical diagnosis

Let D = disease, T = positive test result.

Given: $P(D) = 0.01$ (1% prevalence)
 $P(T|D) = 0.9$ (sensitivity)
 $P(T|\neg D) = 0.1$ (false positive rate)
 Compute $P(D|T)$:

$$P(D|T) = \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.1 \times 0.99} = 0.0089 \approx 0.9\%$$

3. Random Variables and Probability Distributions

- A random variable (RV) is a variable whose value is uncertain.
- A (discrete) probability distribution assigns a probability to each possible value.
- For a random variable X with domain X , $P(X = x) \geq 0$ and $\sum_{x \in X} P(X = x) = 1$.

Example: Weather random variable

Let X = Weather, with possible values {sunny, cloudy, rainy}.

x (Weather,	$P(X = x)$
sunny	0.6
cloudy	0.3
rainy	0.1

This distribution summarizes our beliefs about the weather.

6. Independence and Conditional Independence

- Independence: X and Y are independent if $P(X, Y) = P(X)P(Y)$ or $P(X|Y) = P(X)$.
- Conditional independence: X and Y are conditionally independent given Z if $P(X, Y|Z) = P(X|Z)P(Y|Z)$ (equivalently, $P(X|Y, Z) = P(X|Z)$).

Example: Independence vs. Conditional Independence

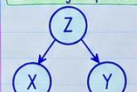
Independent variables

Conditionally independent



$$P(X, Y) = P(X)P(Y)$$

$$X \perp Y$$



$$P(X, Y|Z) = P(X|Z)P(Y|Z)$$

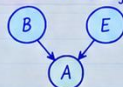
$$X \perp Y | Z$$

Given Z , X and Y become independent (e.g., two sensors independent given the true state).

7. Why Bayesian Networks?

- Full joint distributions are exponentially large (intractable for many variables).
- Many variables are conditionally independent.
- Bayesian networks (BBNs) provide a compact representation using a directed acyclic graph (DAG).
- Each node has a conditional probability distribution (CPD) given its parents.
- Exploits conditional independence to simplify representation and inference.

Example: Simple Bayesian network for the burglary domain



B : Burglary
 E : Earthquake
 A : Alarm

The full joint distribution factorizes as:

$$P(B, E, A) = P(B)P(E)P(A|B, E)$$

Much smaller than specifying $P(B, E, A)$ directly (which needs $2^3 = 8$ numbers).

Advantages:

- Compact representation.
- Encodes conditional independence.
- Supports efficient inference algorithms (exact and approximate).
- Flexible and expressive (works with discrete and continuous variables).
- Widely used in many real-world applications (diagnosis, tracking, decision making).

1. Bayesian Networks Overview

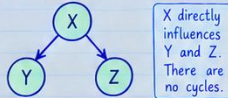
- A Bayesian network (BN) is a probabilistic model that represents a joint distribution over a set of random variables using a directed acyclic graph (DAG).
- Each node is a random variable.
- Arrows represent direct probabilistic influence between variables.
- The network encodes conditional independence assumptions, allowing a compact representation of the joint distribution.

Key Idea: A Bayesian network is a DAG together with a conditional probability table (CPT) for each node.

2. Directed Acyclic Graphs (DAGs)

- A Bayesian network is a directed graph with no directed cycles.
- The graph structure encodes conditional independence relationships.
- If there is an arrow from X to Y, then X is a direct probabilistic influence on Y.
- DAGs allow a topological ordering of the variables consistent with the directions of the arrows.

Example: A simple DAG



3. Nodes, Arrows, and Relationships

- Nodes:** represent random variables (usually directed in simple examples).
- Arrows:** represent direct probabilistic influence (not necessarily causal).
- Parents** $\text{Pa}(X)$: variables with arrows into X.
- Children:** variables that X directly influences.
- Ancestors:** all variables that can reach X via a directed path.
- Descendants:** all variables reachable from X via a directed path.

Example: In the burglary network, Burglary and Earthquake are parents of Alarm. Alarm is a parent of JohnCalls and MaryCalls. Burglary is an ancestor of JohnCalls and MaryCalls.

4. Conditional Probability Tables (CPTs)

- Each node X has a conditional probability table $P(X | \text{Pa}(X))$.
- The CPT specifies a distribution over X for each combination of values of its parents.
- If a node has no parents, the CPT is just a prior distribution $P(X)$.

Example: CPT for Alarm given Burglary and Earthquake

B	E	$P(A = \text{true})$	$P(A = \text{false})$
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

This CPT specifies $P(A | B, E)$ for all combinations of B and E.

5. Factorization of the Joint Distribution

- A Bayesian network defines a joint distribution that factorizes according to the network structure:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Pa}(X_i))$$

- Each variable is independent of its non-descendants given its parents (local Markov property).
- The factorization allows a compact representation using local CPTs instead of a full joint table (which is exponential in the number of variables).

Key Idea: The local Markov property: In a BN, each variable X is conditionally independent of its non-descendants given its parents.

6. Conditional Independence

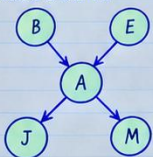
- The graph structure encodes conditional independence assumptions using d-separation (covered in detail later).
- If a path between two variables is blocked given some observed variables, then the variables are conditionally independent.
- Absence of an edge does not mean unconditional independence, but conditional independence given other variables.

Example: In the burglary network, Burglary and Earthquake are independent (unconditionally) because the only path between them is through the collider Alarm.

Notation: $X \perp Y | Z$ means X is conditionally independent of Y given Z.

7. Burglary-Earthquake-Alarm Network

- A classic example from AIMA.
- Variables:
 - B : Burglary (true/false)
 - E : Earthquake (true/false)
 - A : Alarm (true/false)
 - J : JohnCalls (true/false)
 - M : MaryCalls (true/false)
- Structure: B and E influence A, which influences J and M.



This BN has 5 nodes and 4 edges. It encodes many conditional independence relationships.

8. CPTs for the Burglary Network

- Prior probabilities for root nodes:

$$P(B = \text{true}) = 0.001$$

$$P(E = \text{true}) = 0.002$$

- CPT for Alarm:

B	E	$P(A = \text{true})$	$P(A = \text{false})$
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

- CPT for JohnCalls (depends on Alarm):

A	$P(J = \text{true})$	$P(J = \text{false})$
T	0.90	0.10
F	0.05	0.95

- CPT for MaryCalls (depends on Alarm):

A	$P(M = \text{true})$	$P(M = \text{false})$
T	0.70	0.30
F	0.01	0.99

9. D-Separation (Intuition)

- D-separation is a graphical criterion for determining conditional independence from the network structure.
- A path is blocked if:
 - It contains a non-collider that is observed (in Z).
 - It contains a collider ($\rightarrow X \leftarrow$) that is not observed and has no observed descendant.
- If all paths between X and Y are blocked given Z, then $X \perp Y | Z$.

Example: In the burglary network:

- $B \perp E$ (independent).
- $B \perp J | A$ (independent given Alarm).
- $J \perp M | A$ (independent given Alarm).
- B and J are dependent (via A) if A is not observed.

10. Advantages of Bayesian Networks

- Compact representation of large joint distributions.
- Explicitly encodes conditional independence.
- Supports efficient inference algorithms.
- Flexible and expressive for real-world problems (e.g., diagnosis, monitoring, prediction).
- Provides a natural language for reasoning about uncertain domains.

1. Inference in Bayesian Networks

- Inference means computing the probability of a query given some evidence: $P(X|e)$.
- We are given a Bayesian network (BN) that specifies a joint distribution compactly.
- Queries may involve observed variables (evidence) and unobserved variables (hidden variables).
- Exact inference computes the true posterior probability using the BN structure.

Key Idea: Use the graphical structure to compute probabilities more efficiently than enumerating the full joint distribution.

4. Factors in Probabilistic Inference

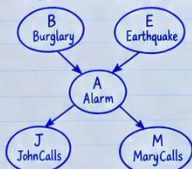
- A factor is a function that maps a set of variables to a non-negative number.
- Factors can represent conditional probability tables (CPTs) or intermediate products during inference.
- Multiplying factors combines information; summing out a variable marginalizes it.
- Factors allow us to manipulate distributions locally without constructing the full joint.

Notation:

- A factor over variables $X_1, \dots, X_n$ is written as $f(X_1, \dots, X_n)$.
- Pointwise product: $(f \times g)(x) = f(x)g(x)$.
- Marginalization (sum-out):
$$\sum_x f(x, y) = g(y).$$

7. Worked Example: Burglary Network Query

Compute: $P(B|J = \text{true}, M = \text{true})$



CPT snippets:

- $P(B=t) = 0.001$
- $P(E=t) = 0.002$
- $P(A|B, E)$
- $P(J|A)$
- $P(M|A)$

Query using enumeration:

$$P(B|j, m) = \alpha \sum_e \sum_a P(B)P(E=e)P(A=a|B, e) \times P(j|a)P(m|a)$$

This sums over the hidden variables A and E. The result is $P(B|j, m) \approx 0.284$ (using the standard CPT values from the book).

2. Queries, Evidence, and Hidden Variables

- Query: the variable(s) whose probability we want to compute (e.g. $P(X|e)$).
- Evidence: observed variables with known values ($e = e_1, \dots, e_k$).
- Hidden variables: unobserved variables that are neither query variables nor evidence. They must be summed out (considered in all possible ways).
- Inference computes the posterior distribution over the query variables given the evidence by eliminating (hidden) variables.

Formulation:

- Let X = query variable(s).
- e = observed evidence variables.
- Y = set of hidden variables.
- We want to compute:

$$P(X|e) = \alpha \sum_y P(X, e, y)$$

(α is a normalization constant)

5. Inference by Enumeration

1. Start with the joint distribution from the BN: $P(X_1, \dots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$.
2. For each query value x , sum over all hidden variables y :

$$P(x, e) = \sum_y P(x, e, y).$$

3. Normalize to get the posterior:

$$P(x|e) = \alpha P(x, e)$$

where $\alpha = \frac{1}{\sum_z P(x, e)}$.

- Simple and always works.
- Time complexity exponential in the number of hidden variables.
- Does not exploit problem structure efficiently.

8. Why Exact Inference Can Be Expensive

- The enumeration algorithm sums over all possible assignments to the hidden variables.
- Time complexity is $O(d^h)$, where:
 - d = maximum number of values per variable
 - h = number of hidden variables (not in query or evidence).
- In many networks, h can be large, leading to exponential cost.
- Even with a compact BN representation, exact inference can be infeasible for large networks (e.g. many variables, high treewidth).

Key Point: Exact inference is tractable for some network structures (e.g. trees or low treewidth), but can be intractable in general.

3. Normalization

- The quantity $P(X, e)$ computed by summing over hidden variables is not normalized over X .
- To get a valid probability distribution, we normalize:

$$P(X|e) = \frac{P(X, e)}{\sum_x P(X, e)} = \alpha P(X, e)$$

where $\alpha = \frac{1}{\sum_x P(X, e)}$.

- Ensures that $\sum_x P(X|e) = 1$.
- Normalization is required for both enumeration and variable elimination.

6. Variable Elimination

- More efficient exact inference algorithm than full enumeration.
- Eliminates hidden variables one at a time by summing them out from factors.
- Works directly with factors, without constructing the full joint distribution.
- The order in which variables are eliminated can greatly affect the computational cost.

Algorithm (Variable Elimination)

1. Build initial factors: one factor per CPT, instantiating evidence variables.
2. While there are hidden variables left:
 - a. Choose a hidden variable Z to eliminate.
 - b. Multiply all factors that contain Z .
 - c. Sum out Z : $f' = \sum_z f$.
 - d. Add the resulting factor back.
3. Multiply remaining factors and normalize.

Example: Factor multiplication and sum-out

Factor 1: $f_1(A, B)$			Factor 2: $f_2(B, C)$			Multiply and sum out B: $f(A, C) = \sum_b f_1(A, b) \times f_2(b, C)$. (Results in a factor over A, C).
A	B	f_1	B	C	f_2	
t	t	0.3	t	t	0.9	
t	f	0.7	t	f	0.1	
f	t	0.2	f	t	0.4	
f	f	0.8	f	f	0.6	

9. Full Joint vs. Structured Inference

- A BN defines the full joint distribution:
 $P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$.
- Exact inference computes $P(X|e)$ using the network structure without enumerating all n variables at once.
- Exploits conditional independences to avoid working with the full joint table, which would have size $O(d^n)$.

10. Key Takeaways

1. Inference answers queries $P(X|e)$ by summing out hidden variables and normalizing.
2. Factors provide a compact way to represent and manipulate distributions.
3. Variable elimination is more efficient than enumeration.
4. The elimination order affects the computational cost.
5. Exact inference can be expensive (exponential in # hidden vars).
6. A BN represents a full joint distribution, but inference uses its structure to be more efficient.

1. Why Approximate Inference?

- Exact inference (variable elimination, junction tree) can be computationally expensive (exponential in treewidth).
- Many real-world problems have large networks with many variables.
- Approximate inference uses sampling to estimate probabilities.
- Trading exactness for efficiency.
- Often gives good estimates with relatively few samples.

Key Idea: Use random sampling to estimate the probability distribution of query variables given evidence.

4. Rejection Sampling

- Generate samples from the prior $P(X)$.
- Keep only samples consistent with evidence E .
- Estimate probabilities using relative frequencies of accepted samples.
- Simple but can be very inefficient if evidence has low probability.

$$P(X = x | E = e) \approx \frac{\# \text{ of samples with } X = x \text{ and } E = e}{\# \text{ of samples with } E = e}$$

Example: Estimate $P(R | S = \text{true})$

- Generate many samples from $P(C, S, R)$.
- Keep only samples where $S = \text{true}$.
- Return fraction of accepted samples with $R = \text{true}$.

7. Markov Chain Monte Carlo Intuition

- Construct a Markov chain whose stationary distribution is the desired posterior $P(X|E)$.
- At each step, make a small random change to the current state (e.g. resample one variable).
- The chain eventually "forgets" the initial state and converges to the target distribution.
- Collect samples after a burn-in period and use them to estimate probabilities.

Intuition: random walk in state space



2. General Sampling Strategy

- Generate samples (complete assignments) from the joint distribution $P(X)$.
- Keep the samples that are consistent with the evidence E (or weight them).
- Estimate $P(X|E)$ using relative frequencies or weighted counts.

Estimate by sampling:

$$P(X = x | E = e) \approx \frac{1}{N} \sum_{i=1}^N f(x, e, s_i)$$

where $f(x, e, s_i)$ depends on the method:

- 1 if sample s_i is consistent with $X = x$ and evidence e (rejection sampling),
- a weight w_i (likelihood weighting), or
- implicitly via a Markov chain (Gibbs sampling).

5. Likelihood Weighting

- Fix evidence variables to their observed values during sampling.
- Sample non-evidence variables from their conditional distributions given parents.
- Each sample is weighted by the likelihood of the evidence given the sampled values.
- More efficient than rejection sampling.

Weight of a sample:

$$w = \prod_{i=1}^k P(E_i = e_i | \text{Parents}(E_i))$$

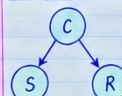
Example: Evidence $S = \text{true}$, $E = \text{false}$

- Fix $S = \text{true}$, $E = \text{false}$.
- Sample other variables (e.g. C , R) from their conditional distributions.
- Compute weight: $w = P(S = \text{true} | C) \times P(E = \text{false} | C, R)$.
- Accumulate weighted counts for query variables (e.g. R).

3. Prior Sampling

- Directly sample complete assignments from the joint distribution $P(X)$ using the network's conditional probabilities.
- Very simple, but ignores evidence during sampling.
- Most samples will be inconsistent with evidence if evidence is unlikely.

Example: Sampling from a simple network



Sampling procedure:

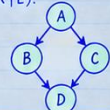
- Sample $C \sim P(C)$
- Sample $S \sim P(S|C)$
- Sample $R \sim P(R|C)$

Repeat to generate complete assignments (C, S, R) .

6. Gibbs Sampling (MCMC)

- A Markov chain Monte Carlo (MCMC) method for approximate inference.
- Start with a complete assignment consistent with evidence.
- Repeatedly sample each non-evidence variable from its Markov blanket given current values of its neighbors.
- After a burn-in period, the samples are approximately from $P(X|E)$.

Markov blanket:
The parents, children, and co-parents of a variable.



Markov blanket of B:
A (parent), D (child),
C (co-parent).

Sampling distribution for variable X_i :

$$P(X_i | \text{MB}(X_i)) \propto P(X_i | \text{Parents}(X_i)) \times \prod_{Y \in \text{Children}(X_i)} P(Y | \text{Parents}(Y))$$

(normalized over all values of X_i)

8. Comparing Sampling Methods

Method	How it works	Pros	Cons
Prior Sampling	Sample from $P(X)$	Very simple	Ignores evidence; very inefficient with unlikely evidence
Rejection Sampling	Sample from $P(X)$ and keep consistent samples	Simple, unbiased	Very inefficient when $P(E)$ is small
Likelihood Weighting	Sample non-evidence vars with weights	More efficient: uses all samples	Weights can have high variance
Gibbs Sampling	MCMC, sample from Markov blankets	Works well for many problems	Need burn-in, autocorrelation, can mix slowly

9. Exact vs. Approximate Inference

Aspect	Exact Inference	Approximate Inference
Result	Exact probabilities	Estimated probabilities
Computation cost	Exponential in treewidth	Typically linear in number of samples
Memory	Can be large	Lower memory (especially sampling)
When to use	Small/moderate networks	Large networks, high treewidth
Examples	Variable elimination, junction tree	Prior, rejection, likelihood weighting, Gibbs sampling

10. Applications and Examples

- Medical diagnosis (large networks).
- Sensor networks and robotics (topic variables).
- Natural language processing (topic models).
- Computer vision (image understanding).
- Genetic networks (bioinformatics).
- Any domain with large probabilistic models where exact inference is infeasible.

"When exact inference is intractable, sampling methods provide a practical way to estimate probabilities."

1. Types of Probabilistic Reasoning

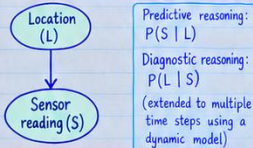
- Diagnostic reasoning (cause from effect)**
 - Infer the cause given an observed effect.
e.g., $P(\text{Disease} \mid \text{Test result})$.
- Predictive reasoning (effect from cause)**
 - Infer the effect given a known cause.
e.g., $P(\text{Test result} \mid \text{Disease})$.
- Intercausal reasoning (explaining away)**
 - Infer between multiple causes of a common effect.
 - Evidence for one cause reduces the probability of other causes.

Key Idea: Bayesian networks support all three types of reasoning naturally through conditional probabilities and graph structure.

4. Robot Localization Example

- A robot is in one of several locations (L).
- Sensor readings (S) provide noisy evidence about the location.
- Use a Bayesian network (often a Markov model) to compute $P(\text{Location} \mid \text{Sensor readings})$.
- This is predictive reasoning (inferring effect of location) and diagnostic reasoning (inferring location from observations).

Example: Robot localization (single time step)



7. Key Formulas Recap

- Bayes' rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- Full joint distribution factorization:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{Parents}(x_i))$$

- Marginalization (sum over hidden variables):

$$P(X) = \sum_Z P(X, Z)$$

- Conditional independence (in a BN):

$$X \perp Y \mid Z \Leftrightarrow P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$$

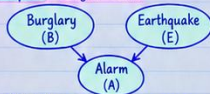
- MAP estimate (most likely explanation):

$$\arg \max_x P(x|e) = \arg \max_x P(X, e)$$

2. Explaining Away (Intercausal)

- When two (or more) possible causes can independently produce the same effect, evidence for one cause reduces the probability of the other.
- The causes become conditionally dependent once we observe the common effect.
- Explains how new evidence can "explain away" a previous hypothesis.

Example: Burglar alarm



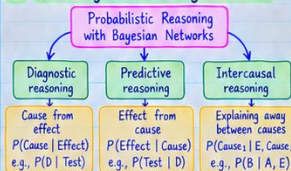
- $P(B|A)$ is high if alarm sounds.
- If we later observe E (earthquake), $P(B|A, E)$ decreases because the earthquake can also explain the alarm (explaining away).

5. Exact vs. Approximate Inference

- Exact inference algorithms:**
 - Enumeration (simple, exponential time).
 - Variable elimination (more efficient).
 - Belief propagation (exact on trees).
- Approximate inference algorithms:**
 - Sampling (e.g., likelihood weighting, Gibbs sampling).
 - Used when networks are large or have many loops.
 - Trade-off: approximation for efficiency.

Method	Exact?	Typical Use	Complexity (roughly)
Enumeration	Yes	Small networks	$O(n)$
Variable elimination	Yes	Medium networks	Exponential in treewidth
Belief propagation	Yes (on trees)	Tree-structured networks	$O(n)$
Sampling (likelihood w.)	No (approx)	Large networks	$O(N)$
Gibbs sampling	No (approx)	Large networks with loops	$O(N)$

8. Summary of Reasoning Patterns



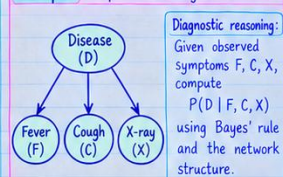
9. Applications in AI

Application	Description	Reasoning Type
Medical diagnosis	Infer disease from symptoms and tests	Diagnostic
Robot localization	Infer location from sensor readings	Diagnostic & Predictive
Fault diagnosis	Find broken component from observed behavior	Diagnostic
Monitoring systems	Predict future state from current evidence	Predictive
Plan recognition	Infer agent's goal from observed actions	Diagnostic

3. Medical Diagnosis Example

- Diseases cause observable symptoms.
- Use probabilistic inference to compute $P(\text{Disease} \mid \text{Symptoms})$.
- Typically a diagnostic reasoning task.

Example: Simple medical diagnosis network



6. Advantages and Limitations of Bayesian Networks

Advantages:

- Compact, structured representation.
- Encodes conditional independence.
- Supports efficient inference algorithms (exact and approximate).
- Natural way to combine prior knowledge with data.
- Works well for uncertain, noisy, real-world domains (e.g., diagnosis, robotics).

Limitations:

- Learning structure and parameters can be challenging (especially for large domains).
- Exact inference can be computationally expensive (NP-hard).
- Requires probabilistic assumptions and the right independence structure.
- May need approximate inference for large or highly connected networks.

10. Exam-Style Questions

- Define diagnostic, predictive, and intercausal reasoning with examples.
- Explain the explaining away phenomenon in a Bayesian network.
- A burglar alarm network: If the alarm sounds and you know there was an earthquake, what happens to $P(\text{Burglary})$?
- Draw a simple Bayesian network for medical diagnosis and write the joint distribution.
- What is the difference between exact and approximate inference? Give examples.
- Explain how Bayesian networks help in robot localization.
- List advantages and limitations of Bayesian networks.
- Write the formula for the MAP estimate and explain its use.

Key Takeaways

- Bayesian networks provide a powerful framework for reasoning under uncertainty.
- They support diagnostic, predictive, and intercausal reasoning.
- Applicable to many real-world AI problems.
- Exact inference is tractable for small/structured networks; approximate inference scales better.
- Understanding probabilistic reasoning is essential for intelligent AI systems.

"Probabilistic reasoning lets us make rational decisions under uncertainty, combining what we know with what we observe." —AIMA (Ch.14)